CSE 270 Sports Analytics

Homework 3

Name: _____

Competitive Balance

Problem 1 (50 points)

Noll-Scully number

1. Get the final tables for all countries in the dataset *f_data_sm* for all seasons(for each country). (10 points)

```
head(table)
```

```
##
## 1 function (..., exclude = if (useNA == "no") c(NA, NaN), useNA = c("no",
## 2     "ifany", "always"), dnn = list.names(...), deparse.level = 1)
## 3 {
## 4     list.names <- function(...) {
## 5         l <- as.list(substitute(list(...)))[-1L]
## 6         if (length(l) == 1L && is.list(..1) && !is.null(nm <- names(..1)))
```

```
table_list = data.frame()

all__countries_seasons<- function(f_data_sm){
  for(c in unique(f_data_sm$COUNTRY)){
    for(s in unique((f_data_sm %>% filter(COUNTRY == c))$SEASON)){
      new_country = final_table(f_data_sm, c, s)
      new_country$SEASON = s
      new_country$COUNTRY = c
      table_list = rbind(table_list, new_country)
    }
  }
  return(table_list)
}
table_list<-all__countries_seasons(f_data_sm)
head(table_list)
```

```
##         TEAM  M  W  D  L GF GA DIFF POINTS POSITION SEASON COUNTRY
## 1 Man United 42 27 11  4 80 38   42     92        1   1994 England
## 2  Blackburn 42 25  9  8 63 36   27     84        2   1994 England
## 3  Newcastle 42 23  8 11 82 41   41     77        3   1994 England
## 4    Arsenal 42 18 17  7 53 28   25     71        4   1994 England
```

```
## 5      Leeds 42 18 16  8 65 39   26     70        5   1994 England
## 6  Wimbledon 42 18 11 13 56 53    3     65        6   1994 England
```

2. Calculate the wining percentage(ratio) for all the teams in all seasons. (Consider draws as half wins). (2 points)

```
table_list <- table_list %>% group_by(COUNTRY, SEASON) %>%
  mutate(wpct = (W + D*0.5) / (W+L+D)) %>% ungroup()
head(table_list)
```

```
## # A tibble: 6 x 13
##   TEAM       M     W     D     L    GF    GA  DIFF POINTS POSIT~1 SEASON COUNTRY
##   <chr>  <dbl> <int> <int> <int> <dbl> <dbl> <dbl>  <dbl>   <int>  <dbl> <chr>
## 1 Man U~    42    27    11     4    80    38    42     92       1   1994 England
## 2 Black~    42    25     9     8    63    36    27     84       2   1994 England
## 3 Newca~    42    23     8    11    82    41    41     77       3   1994 England
## 4 Arsen~    42    18    17     7    53    28    25     71       4   1994 England
## 5 Leeds     42    18    16     8    65    39    26     70       5   1994 England
## 6 Wimbl~    42    18    11    13    56    53     3     65       6   1994 England
## # ... with 1 more variable: wpct <dbl>, and abbreviated variable name
## #   1: POSITION
```

3. Calculate the Noll-Scully number for each season and country using the winning ratio. (3 points)

```
N_S <- table_list %>% group_by(SEASON, COUNTRY) %>%
  summarise(NS=sd(wpct)/(0.5/sqrt(W+L+D)))%>%
  filter(row_number() == 1)
head(N_S)
```

```
## # A tibble: 6 x 3
## # Groups:   SEASON, COUNTRY [6]
##   SEASON COUNTRY        NS
##    <dbl> <chr>       <dbl>
## 1   1994 England      1.56
## 2   1994 France       1.54
## 3   1994 Germany      1.17
## 4   1994 Italy        1.59
## 5   1994 Netherlands  1.61
## 6   1994 Spain        1.33
```

4. Visualize the results grouped by seasons and countries. Make sure to include meaningful title and axis names.
   (You can use facets) (10 points)

5.Interpret the visualization (5 points)

C5 competitive balance

6.Calculate the C5 index for the number of goals scored by the top 5 teams in each country for each season. You should use the dataset from Problem 1.1. (5 points)

7. Visualize the results for all countries and seasons. (Can be the same structure as in 1.4) (10 points)
8. Interpret the results (5 points)

Individual Performance (50 points)

Problem 2

1. Load the dataset *nba_players* from the library *SportsAnalytics270* and filter it to only include seasons starting from 1992.
   Remove the columns blanl and blank2 and drop missing values or nas. (2 points)

2. Clean the variable Pos from players with double positions, for example C-PF should be turned into C. (5 points)

3. Convert the vaiable Pos into a factor and remove the columns X, Year, Player and Tm. (3 points)

4. Divide the dataset into train and test datasets. (You can use dplyr or any other sampling method) (10 points)

5. Build a classifier model to predict the position of the player based on the playing attributes.
   Use the training set for fitting the model. (20 points) You can use multinomial regression for that

6. Predict the position of the players for the test dataset. (5 points)

7. Interpret the predictive power of the model. (5 points)